



MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY, JAMSHORO
DEPARTMENT OF ELECTRONIC ENGINEERING
INTEGRATED ELECTRONICS # 09
To understand the modes and applications of 555 timer IC



Roll No:	Date:
Checked by:	Score:

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	DESIGN AND CODING	OBSERVATION/ RESULTS	SCORE

Performance:

Objective: To be familiar with the modes of 555 as Timer & Multivibrator, Application.

Apparatus:

- i. 555 Timer IC
- ii. Resistors
- iii. Capacitor
- iv. Oscilloscope
- v. Function Generator
- vi. DC-voltage power supply

Theoretical Background:

Introduction:

The 555 IC is unique in that it simply, cheaply, and accurately serves as a free-running astable multivibrator, square-wave generator, or signal source, as well as being useful as a pulse generator and serving as a solution to many special problems. It can be connected as a one shot or an astable oscillator, as well as being used for a multitude of custom designs. Figure 7.1 shows a block diagram of the chip with its internal components. The 555 timer got its name from the three 5-k Ω resistors. They are set up as a voltage divider from VCC to ground. The top of the lower 5 k Ω is at $1/3$ VCC and the top of the middle 5 k Ω is at $2/3$ VCC.

For example, if VCC is 6V, each resistor will drop 2V. The triangle shaped symbols represents comparators. The S-R flip-flop is driven by the two comparators. It has an active-LOW Reset and its output is taken from the Q. The discharge transistor is an NPN, which is used to short pins 7 to 1 when Q is HIGH.

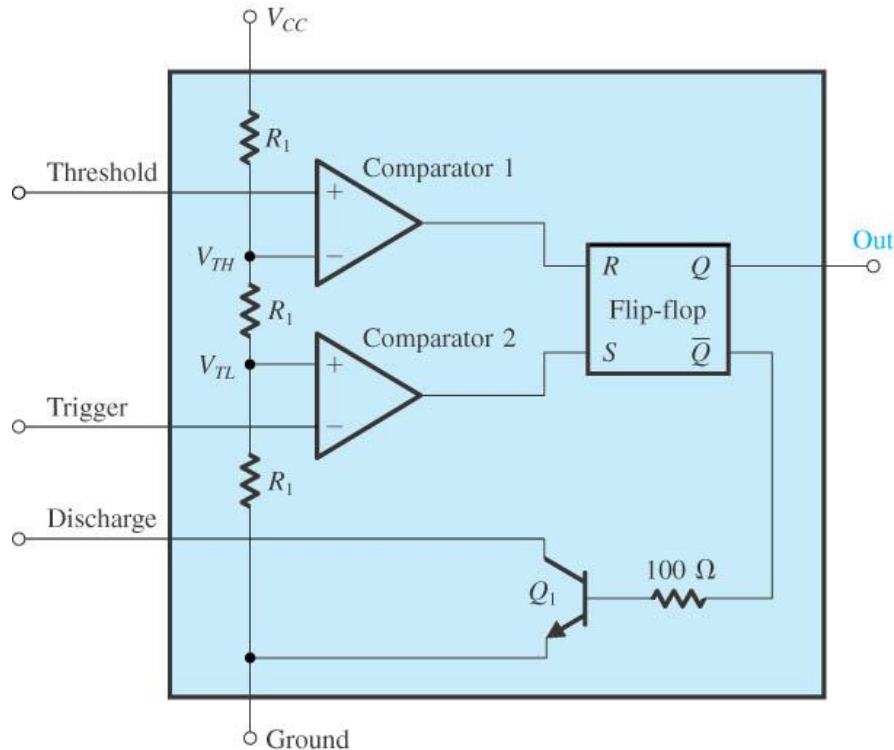
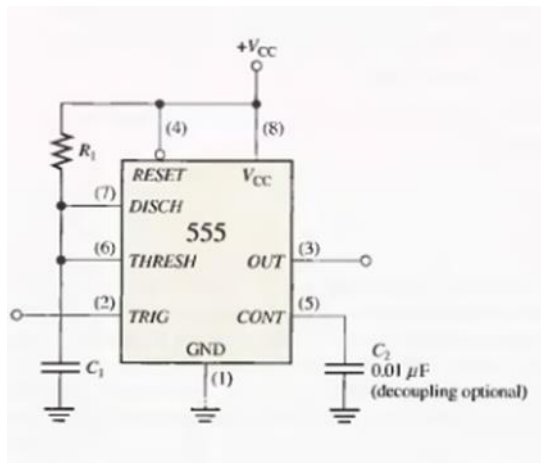


Figure 7.1: Block Diagram of 555 Timer

Monostable (One shot) Operation:

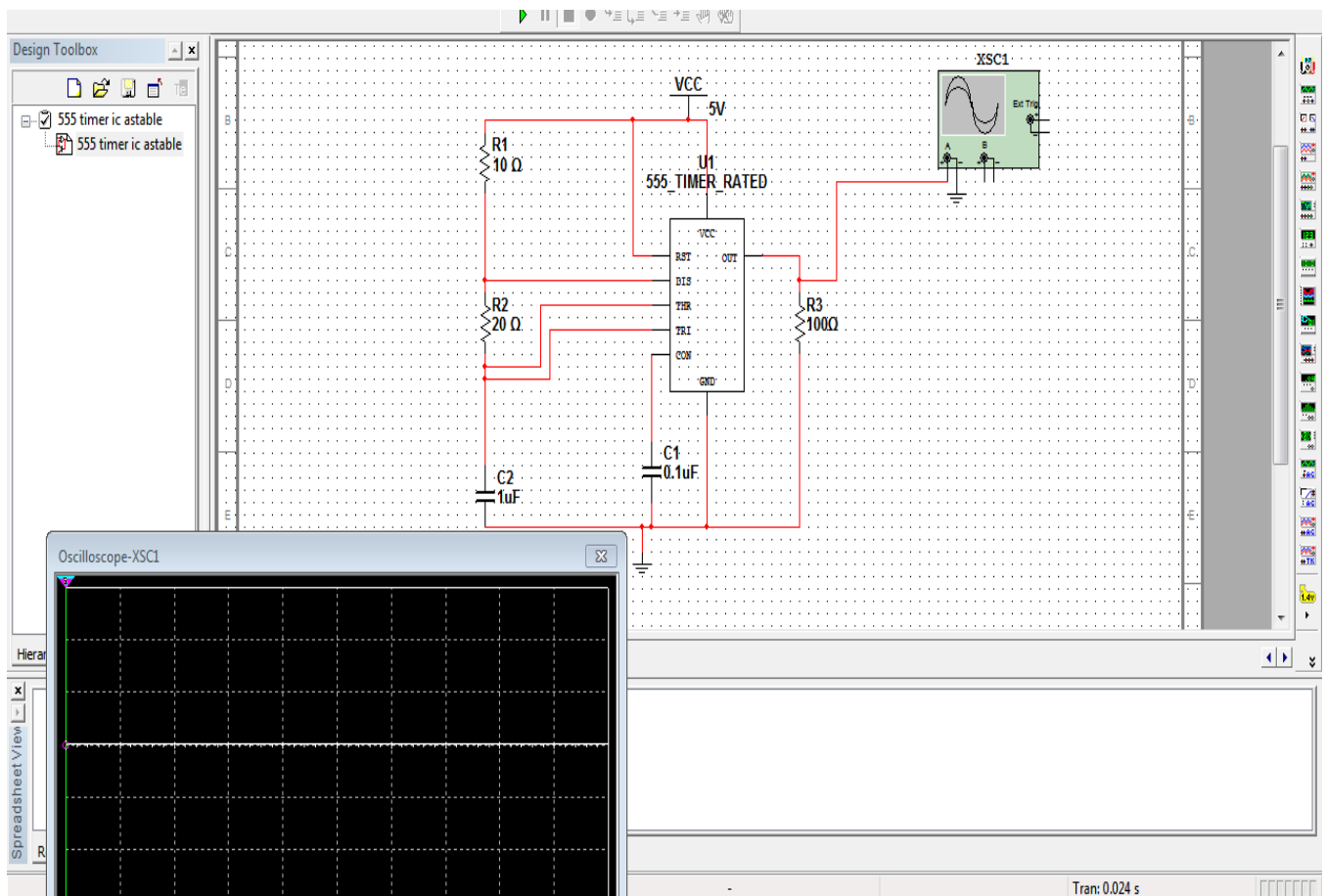
The Monostable circuit outputs one pulse for each high to low transition of the trigger pin. The output will return to the “stable” state after a period of time that is a function of the attached resistor and capacitor values.

The discharge pin is normally connected to ground (internally). When the trigger pin transitions from V_{CC} to 33% V_{CC} the discharge pin is then internally disconnected from ground and the output pin is set high. The capacitor C then starts to charge through resistor R . The threshold pin detects when the voltage across the capacitor reaches 66.6% V_{CC} . When the voltage across the capacitor reaches 66.6% V_{CC} , the output pin is set low and the discharge pin is connected back to ground. Finally, when the discharge pin is connected back to ground, the capacitor is discharged and the default “stable” state is reached.



Activity: 20 sec delay $c = 200 \text{ mF}$ $R = 91 \text{ k ohm}$

1. Design a circuit that switch ON/OFF a DC motor after every 20 seconds

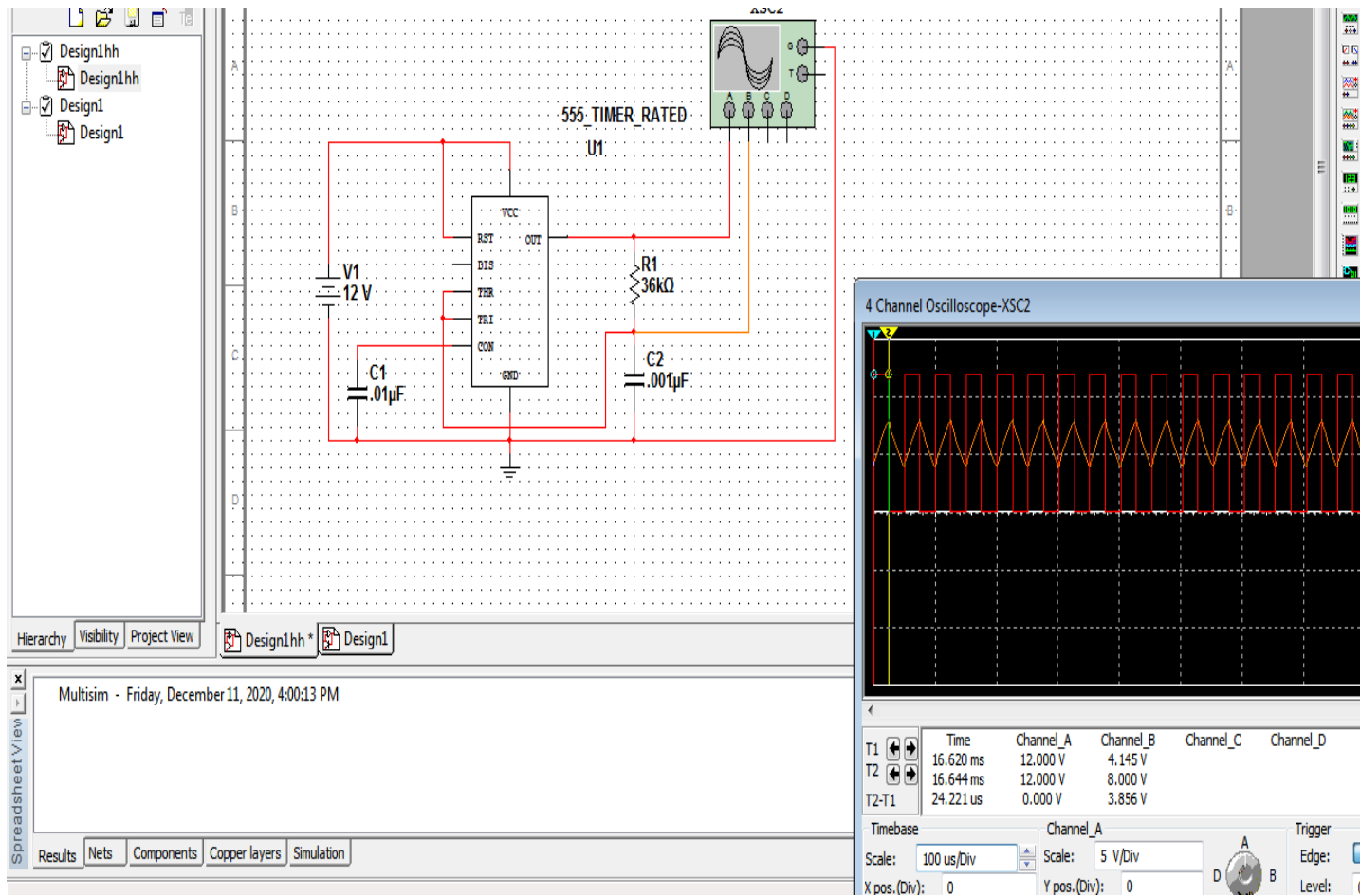


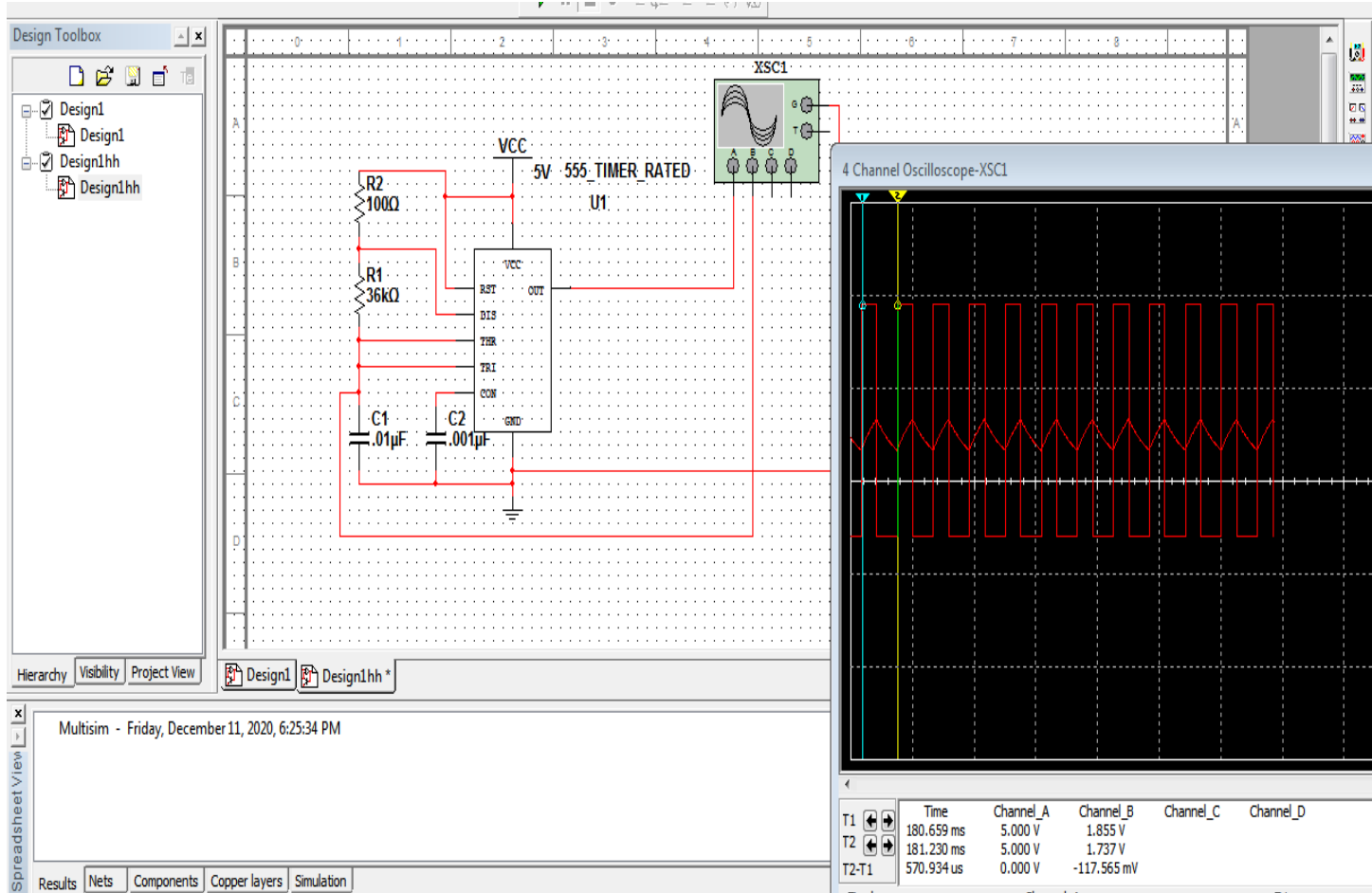
2. Design a circuit with 555 Timer that produces the 50% duty cycle. Activate the transistor when the output is high. Enabling of transistor must saturate the high alarm comparator with reference voltage $(2+0.5 \times R)V$ [Where R is the LSB of your roll number] that drives any load through a relay. [Hint: Select the transistor output by own which could drive/activate the comparator.]

Design a timer circuit with $t_H = (50+R)ms$ and $t_L = (40+R)ms$ [Where R is your roll number least significant bit]. Choose the suitable capacitor value in between the 0.1 to $1\mu F$ and calculate the resistor values. Simulate the circuit in multisim and observe the results whether matches your calculation

Task no 2(circuit using 555 timer that produces 50% duty cycle here we calculate T_H and T_L values and set the resistor and capacitor values according to that and hence

achieve the 50% duty cycle





timer circuit with $t_H = (50 + R)ms$ and $t_L = (40 + R)ms$ here calculated resistors values are from the formula of time high and time low

here CALCULATED VALUE OF $R1 = 20 \text{ OHM}$, $R2 = 95 \text{ OHM}$, $C = 0.7 \text{ UF}$

so by these values we design a timer circuit

Digital Oscilloscope

